Beyond Visual Quality: Evaluating the Robustness and Recognition Performance of Image Cartoonization Methods

Research project summary, current findings, literature positioning, and future extensions

Purpose: A consolidated summary for supervisor discussion and project documentation. This report describes completed experiments and distinguishes them from proposed future work.

Author: Karthikeya Satya Sai Pavan Challapalli

Affiliation: Department of Computer Applications, Vignan's University (confirm the official institutional wording before submission)

Status: Experimental results summarized; novelty is not certified; further verification is needed before journal submission.

1. Executive summary

This project compares three cartoonization pipelines—OpenCV cartoonization, pretrained AnimeGANv2, and a pretrained Stable Diffusion image-to-image pipeline—using a fixed ResNet18 classifier. The dataset is a balanced subset of 300 Imagenette images across ten classes, with a recorded selection seed of 42. The study measures clean-image recognition, performance under four selected degradations, reference-image similarity, per-class outcomes, prediction consistency, statistical comparisons, confusion matrices, and visual failure cases.

The strongest descriptive finding is that AnimeGANv2 had the highest recognition accuracy among the three cartoonization methods in the current experimental setup. This does not establish that it is universally the best cartoonization method; conclusions are limited to the current sample, classifier, checkpoints, and settings.

2. Experimental workflow

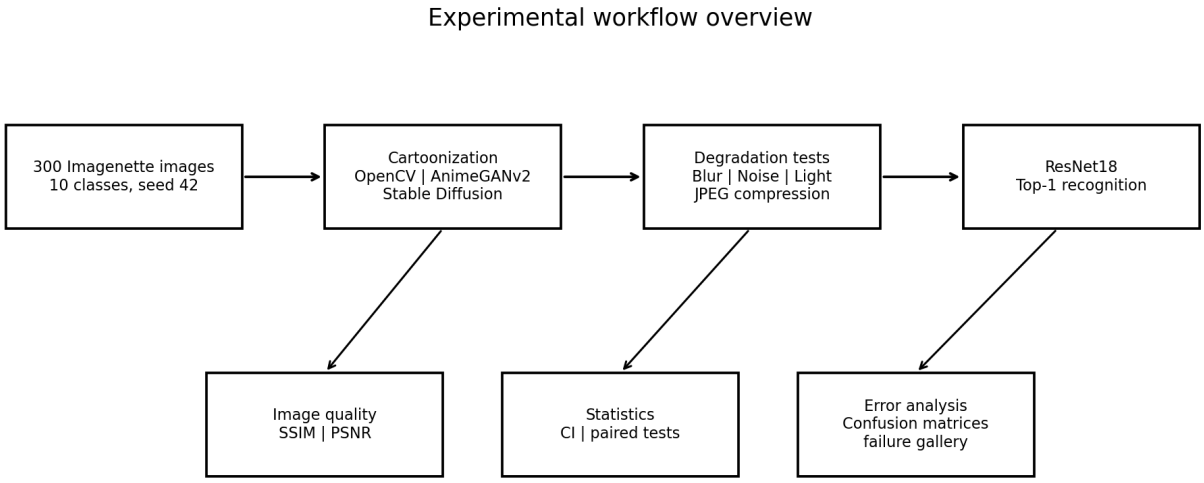


Figure 1. Workflow diagram generated from the recorded experimental design. Exact model settings should be included in the reproducibility archive.

3. Clean-image recognition results

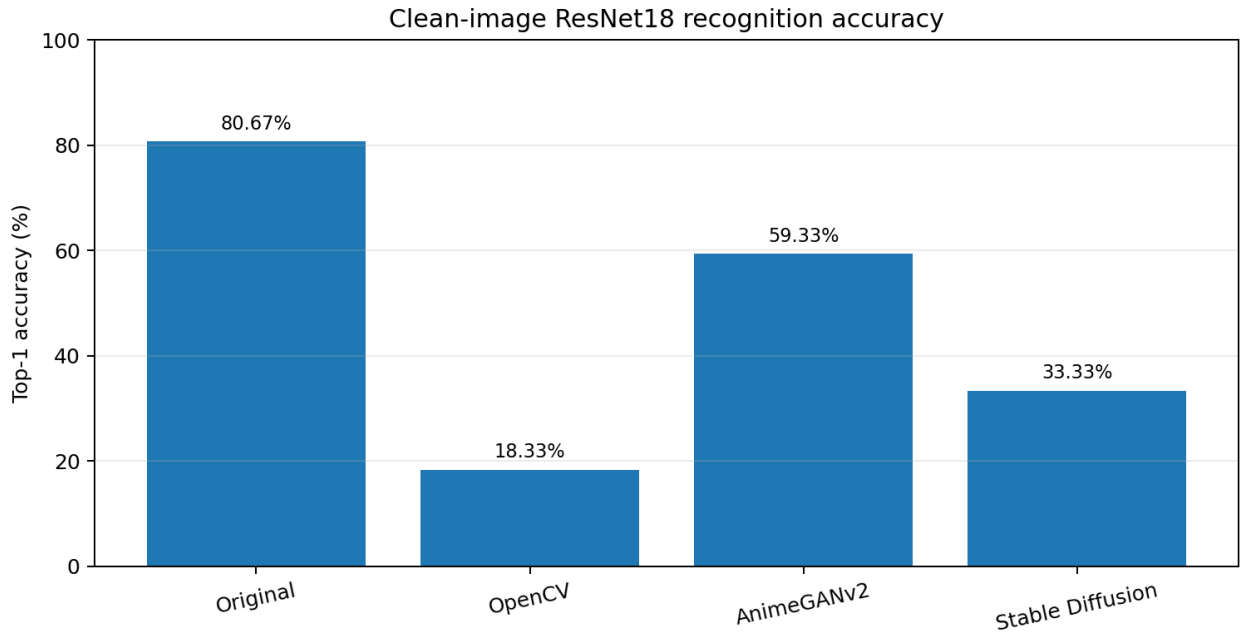


Figure 2. ResNet18 top-1 accuracy on clean images. Stable Diffusion uses 294 valid outputs after excluding six black outputs.

Method	Samples	Correct	Accuracy
Original	300	242	80.67%
OpenCV	300	55	18.33%
AnimeGANv2	300	178	59.33%
Stable Diffusion	294	98	33.33%

Interpretation: the original images achieved 80.67% accuracy. Among the cartoonized outputs, AnimeGANv2 achieved 59.33%, Stable Diffusion 33.33%, and OpenCV 18.33%. These are observed results for this sample, not universal rankings.

4. Accuracy under selected degradation conditions

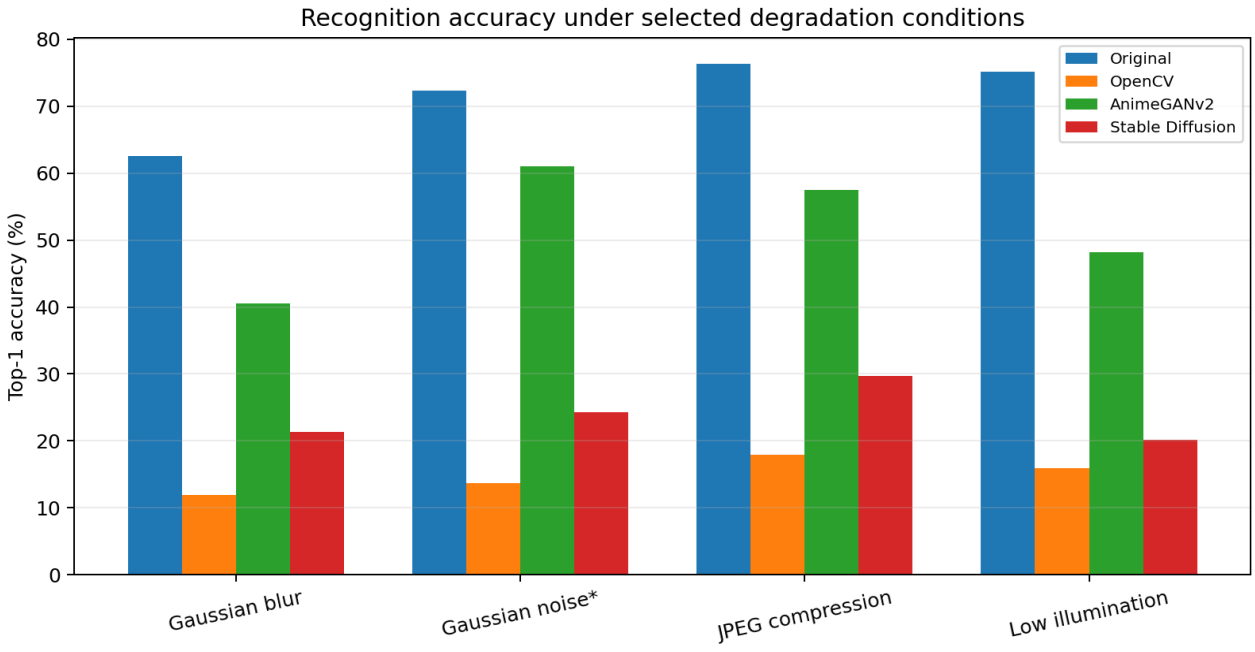


Figure 3. Reported accuracy under Gaussian blur, Gaussian noise, JPEG compression, and low illumination.

Method	Gaussian blur	Gaussian noise*	JPEG compression	Low illumination
Original	62.59%	72.33%	76.35%	75.18%
OpenCV	11.89%	13.67%	17.91%	15.96%
AnimeGANv2	40.56%	61.00%	57.43%	48.23%
Stable Diffusion	21.33%	24.24%	29.73%	20.21%

Important limitation: The Gaussian-noise summary was not calculated using exactly the same matched image IDs as the other degradation comparisons. Treat the noise comparison as provisional until it is recalculated on identical samples using saved predictions. The other degradation summaries also have matched-subset sample counts below 300 because invalid Stable Diffusion outputs are excluded.

5. Reference-image quality metrics

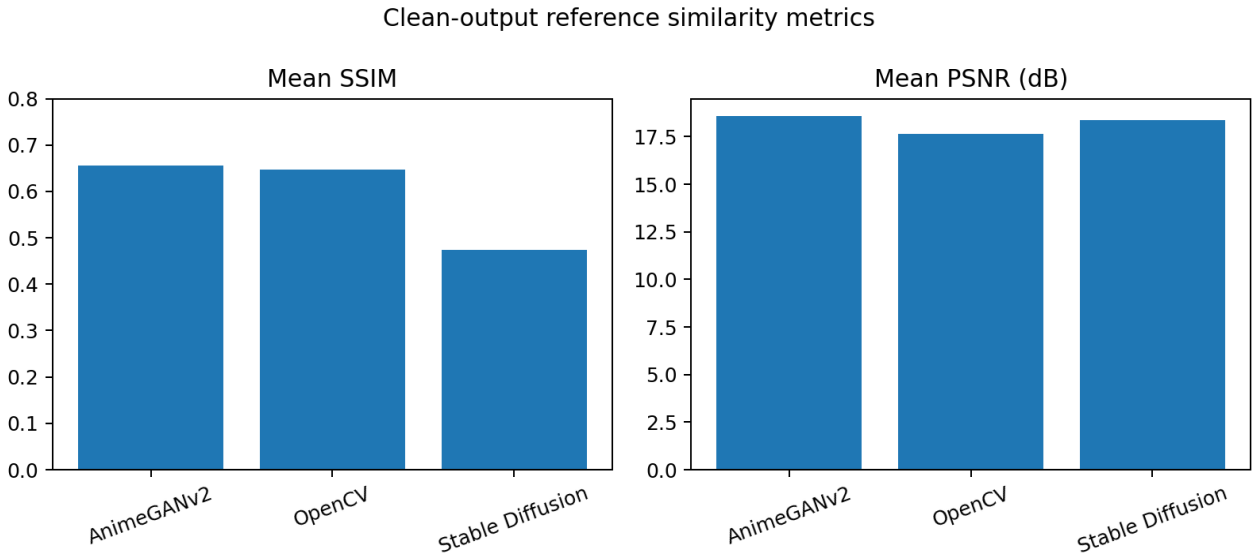


Figure 4. Mean SSIM and PSNR between original images and clean cartoonized outputs.

Method	Valid images	Mean SSIM	Mean PSNR (dB)
AnimeGANv2	300	0.6564	18.5636
OpenCV	300	0.6469	17.6617
Stable Diffusion	294	0.4747	18.3549

AnimeGANv2 had the highest reported mean SSIM and PSNR. SSIM and PSNR measure reference-image similarity; they do not directly measure aesthetic quality, semantic correctness, or human preference.

6. Other analyses completed

Prediction consistency: saved comparison of how often classifier predictions remain unchanged between original and cartoonized images. Consistency is not the same as correctness.

Per-class analysis: AnimeGANv2 had the highest reported accuracy in nine of ten categories; Stable Diffusion slightly exceeded it for garbage truck (60.00% versus 56.67%). Stable Diffusion denominators varied because some outputs were invalid or unavailable.

Failure-case gallery: 12 examples were selected where the original image was correctly classified but at least one cartoonized output was misclassified. Three gallery images were generated.

Confusion matrices: four matrices were generated, with predictions outside the ten selected categories grouped as “Other classes.” The corrected overall accuracies agree with the main clean-image table.

Statistics: Wilson confidence intervals, paired McNemar tests with Bonferroni correction, and paired bootstrap intervals were saved. Verify exact p-values, matched IDs, and interval directions from the CSVs before publication.

7. Literature review and research-gap positioning

Relevant prior work includes CartoonGAN (Chen et al., CVPR 2018), white-box cartoonization (Wang & Yu, CVPR 2020), ImageNet-Cartoon/ImageNet-Drawing (Salvador & Oberman, ICML Shift Happens workshop 2022), ToonNet (VRCAI 2018), stylization and shape/texture bias (Geirhos et al., ICLR 2019), and common-corruption robustness (Hendrycks & Dietterich, ICLR 2019). These works mean the broad ideas of cartoonization, cartoon-domain classification, stylization-related recognition, and corruption robustness are not new.

A safe contribution statement is: “This study presents a controlled empirical comparison of OpenCV cartoonization, pretrained AnimeGANv2, and a pretrained Stable Diffusion image-to-image pipeline under a fixed ResNet18 classifier. It evaluates clean inputs and selected common corruptions, and reports recognition accuracy, reference-image similarity, per-class outcomes, and failure cases on a balanced 300-image subset of ten Imagenette classes.”

Novelty status: not certified. This report is a targeted narrative summary, not an exhaustive systematic review. Do not claim “first study” or “no prior research exists” until the closest studies have been searched and compared in full text across scholarly databases.

8. Future extensions

Priority 1 — Matched-sample analysis. Recalculate Gaussian-noise results on identical image IDs across all methods using existing per-image prediction CSVs wherever possible. This is the first required correction and does not require rerunning cartoonization models.

Priority 2 — Multiple corruption severity levels. Evaluate several strengths of Gaussian blur, Gaussian noise, low illumination, and JPEG compression. Apply degradations to saved images and run classifier inference; generative-model reruns are unnecessary for existing images.

Priority 3 — Larger sample or second dataset. If compute allows, expand the balanced sample or validate on a second dataset. New images need corresponding cartoon outputs from all methods, so this extension has a higher compute cost.

Priority 4 — Second classifier. Evaluate ResNet50 or another established classifier on the existing original and cartoonized outputs to test whether the ranking depends on ResNet18. This requires classifier inference, not cartoonization reruns.

9. Recommended finalization order

1. Recompute matched-sample Gaussian-noise metrics. 2. Verify paired statistics, p-values, confidence intervals, and output exclusions. 3. Record exact model versions, prompts, seeds, parameters, software, and hardware. 4. Finish a documented literature search and compare the closest papers. 5. Decide with the supervisor whether severity levels, more data, or a second classifier are feasible. 6. Finalize the manuscript and format it for the target venue.

10. Reproducibility checklist

Preserve the executed Colab notebook (.ipynb) and a code export.

Include selected_300_images.csv and document the dataset source/version and license.

Include source-to-output mappings, generated images, all prediction CSVs, quality metrics, statistical outputs, graphs, confusion matrices, and failure-case files.

Document all model checkpoints/revisions, prompts, random seeds, preprocessing and postprocessing.

State the black-output exclusion rule and valid sample count for every analysis.

Do not report runtime/memory comparisons unless comparable measurements exist.

Verify every number in the final manuscript against the saved CSVs.

11. Selected references

[1] Chen, Y., Lai, Y.-K., & Liu, Y.-J. (2018). CartoonGAN: Generative Adversarial Networks for Photo Cartoonization. CVPR. https://openaccess.thecvf.com/content_cvpr_2018/html/Chen_CartoonGAN_Generative_Adversarial_CVPR_2018_paper.html

[2] Wang, X., & Yu, J. (2020). Learning to Cartoonize Using White-Box Cartoon Representations. CVPR. https://openaccess.thecvf.com/content_CVPR_2020/html/Wang_Learning_to_Cartoonize_Using_White-Box_Cartoon_Representations_CVPR_2020_paper.html

[3] Salvador, T., & Oberman, A. (2022). ImageNet-Cartoon and ImageNet-Drawing: Two domain shift datasets for ImageNet. ICML Workshop on Shift Happens. https://openreview.net/pdf?id=YlAUXhjwaQt

[4] Zhou, Y. et al. (2018). ToonNet: A cartoon image dataset and a DNN-based semantic classification system. VRCAI '18. https://doi.org/10.1145/3284398.3284403

[5] Geirhos, R. et al. (2019). ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness. ICLR. https://arxiv.org/abs/1811.12231

[6] Hendrycks, D., & Dietterich, T. (2019). Benchmarking Neural Network Robustness to Common Corruptions and Perturbations. ICLR. https://arxiv.org/abs/1903.12261

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[8] Wang, Z. et al. (2004). Image Quality Assessment: From Error Visibility to Structural Similarity. IEEE Transactions on Image Processing, 13(4), 600–612. https://doi.org/10.1109/TIP.2003.819861

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